

String Operations

TIP101

Agenda

1 Welcome 🙌

2 String Operations

3 **BREAK**

4 Collaborative Problem Solving

5 Wrap Up

10 minutes

35 minutes

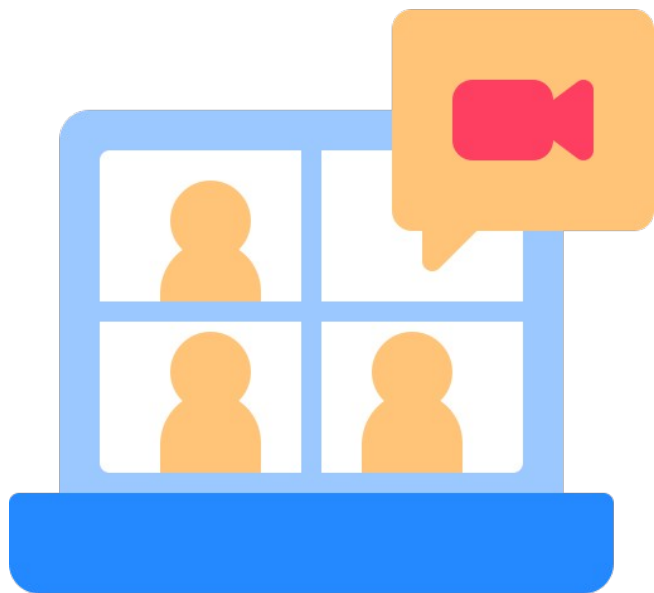
5 minutes

60 minutes

10 minutes

10 minutes

Welcome! 🖐️



**Turn on your
cameras**



**Fill out the entry
ticket**



Icebreaker

5 minutes



Question:

What is one game you always feel confident about winning? (board game, card game, part game, video game, etc?).



Answer:

Share your answer in the Zoom chat below. Multiple options are encouraged.

Your Feedback



Glow

- + Students report feeling supported by classmates during breakout rooms
- + Dictionaries went well



Grow

- Issues with visibility with screen sharing code (zoom in, try VS Code Live)
- Wanted more details on built in functionalities.



Announcements & Reminders

* Assessment Reminder:

- Fix it garage sessions ended for now (onboarding specific to errors relating to VS Code, the course portal, Slack, etc.)
- Study hall sessions will remain active !
- Utilize VS Code debugging functionalities.

35 minutes

Strings in Python



Discussion

5 minutes

Open the [Unit 3 Cheat Sheet](#) and find **String Methods & Syntax**.

What do you notice?

What do you wonder?

String Methods & Syntax

!! This material is in scope for the Unit 3 assessment.

Lower Method

`s.lower()` Returns `s` as a lowercase string. [Try it](#)

- Accepts no parameters
- Returns a lowercase string.

Example Usage:

```
# Example 1: Mixed case
s = 'Hello World!'
lowered = s.lower()
print(lowered) # Prints 'hello world!'

# Example 2: All uppercase
s = 'HELLO WORLD'
lowered = s.lower()
print(lowered) # Prints 'hello world'
```

String Basics

- * Strings are sequences of characters.
- * Common operations: concatenation, slicing, and formatting.
- * Useful methods: `.lower()`, `.upper()`, `.split()`.

```
s = "Hello World"
print(s.lower())  # "hello world"
print(s[:5])     # "Hello"
```

String Concatenation

- * Strings can be built by simply adding them together. This allows you to convert or modify existing text (filtering, encrypting, etc.)
- * Concatenation refers to using the “+” modifier to combine two or more strings together.

```
a = "Hello"  
b = "World"  
c = a + " [Middle Segment] " + b
```

String Slicing

- * Returns a snippet or portion of a string, based on the provided indices.
- * Can provide either the beginning, or the ending index.
 - `value = exampleString[:5]`
 - `value = exampleString[4:]`
 - `value = exampleString[2:8]`
 - `value = exampleString[0:10:2]` #optional "step" value.



Discussion

2 minutes

What do you think happens if we try to modify a string in place? Please answer in the Zoom.

```
# Which section below is correct?
```

```
# Option A
```

```
s = "Hello World"
```

```
s = s.upper()
```

```
# Option B
```

```
s = "Hello World"
```

```
s.upper()
```



Discussion

2 minutes

What do you think happens if we try to modify a string in place? Please answer in the Zoom.

```
# Which section below is correct?  
# Option A ( CORRECT )  
s = "Hello World"  
s = s.upper()  
  
# Option B  
s = "Hello World"  
s.upper()
```



Instructor Demo

10 minutes

Let's explore . . .

- * How to retrieve specific characters or substrings
- * Common transformations (e.g., `upper()`, `replace()`, `split()`)



Instructor Demo

A pangram is a sentence that contains every letter of the English alphabet at least once, regardless of case. Write a function to determine if a given sentence is a pangram.

```
def is_pangram(sentence):  
    pass
```

Input: sentence = "The quick brown fox jumps over the lazy dog"

Output: True

Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.



60 minutes

Collaborative Problem Solving



Breakout Rooms

60 minutes

1. 🏃 Navigate to your breakout room.
2. 🙌 Introduce yourselves!
3. 🧑💻 Work on the problems for your selected unit.

10 minutes

Wrap Up



One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!



Session Survey Time!





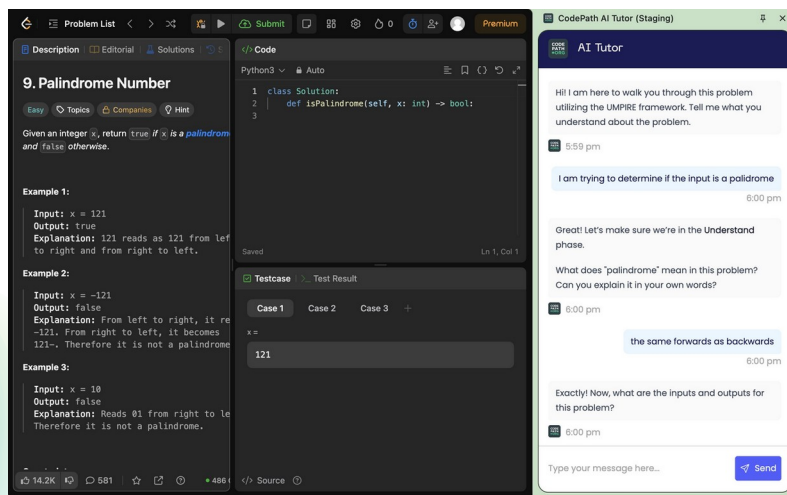
Final Reminders

- ☐ No big reminders, remaining time will be for **Q&A**
- ☐ Take the weekly HackerRank assessment by Sunday @ midnight
- ☐ Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!
- ☐ Remember to check out the **Codepath AI Tutor Bot** and earn an absence back! 

How do I get the most out of this course?



CodePath AI Tutor



Build your skills with support! Take the following steps:

- * Install the AI Tutor on chrome: <http://go.codepath.org/tiptutor>
- * Solve 2 LeetCode problems on 2 different days

Once you've taken the steps above, unlock a new opportunity!

- * +1 absence back for completion(one-time offer)

[Take a look at the FAQ if you have any questions](#)